

gilmore-lie-groups: table evidence

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																												
gilmore-lie-groups_ TAB_001	26	0.877	$\begin{tabular}{t}{ l l l l l }\hline & I & (123) & (321) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & v_1 = s_1 + s_2 + s_3 \\ \Gamma^2 & 1 & \omega & \omega^2 & v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ \Gamma^3 & 1 & \omega^2 & \omega & v_3 = s_1 + \omega^2 s_2 + \omega s_3 \\ \hline\end{tabular}$	<table><tr><td></td><td>I</td><td>(123)</td><td>(321)</td><td>Basis functions</td></tr><tr><td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>$v_1 = s_1 + s_2 + s_3$</td></tr><tr><td>Γ^2</td><td>1</td><td>ω</td><td>ω^2</td><td>$v_2 = s_1 + \omega s_2 + \omega^2 s_3$</td></tr><tr><td>$\Gamma^3$</td><td>1</td><td>$\omega^2$</td><td>$\omega$</td><td>$v_3 = s_1 + \omega^2 s_2 + \omega s_3$</td></tr></table>		I	(123)	(321)	Basis functions	Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$	Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$	Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$	<table><tr><td></td><td>I</td><td>(123)</td><td>(321)</td><td>Basis functions</td></tr><tr><td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>$v_1 = s_1 + s_2 + s_3$</td></tr><tr><td>Γ^2</td><td>1</td><td>ω</td><td>ω^2</td><td>$v_2 = s_1 + \omega s_2 + \omega^2 s_3$</td></tr><tr><td>$\Gamma^3$</td><td>1</td><td>$\omega^2$</td><td>$\omega$</td><td>$v_3 = s_1 + \omega^2 s_2 + \omega s_3$</td></tr></table>		I	(123)	(321)	Basis functions	Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$	Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$	Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$																				
	I	(123)	(321)	Basis functions																																																													
Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$																																																													
Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$																																																													
Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$																																																													
	I	(123)	(321)	Basis functions																																																													
Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$																																																													
Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$																																																													
Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$																																																													
gilmore-lie-groups_ TAB_002	30	0.669	$\begin{tabular}{t}{ l l l l l l }\hline & (I\backslash) & (12)(34) & (13)(24) & (14)(23) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & 1 & w_1 = t_1 + t_2 + t_3 + t_4 \\ \Gamma^2 & 1 & 1 & -1 & -1 & w_2 = t_1 + t_2 - t_3 - t_4 \\ \Gamma^3 & 1 & -1 & 1 & -1 & w_3 = t_1 - t_2 + t_3 - t_4 \\ \Gamma^4 & 1 & -1 & -1 & 1 & w_4 = t_1 - t_2 - t_3 + t_4 \\ \hline\end{tabular}$	<table><tr><td></td><td>I</td><td>(12)(34)</td><td>(13)(24)</td><td>(14)(23)</td><td>Basis functions</td></tr><tr><td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>$w_1 = t_1 + t_2 + t_3 + t_4$</td></tr><tr><td>$\Gamma^2$</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>$w_2 = t_1 + t_2 - t_3 - t_4$</td></tr><tr><td>$\Gamma^3$</td><td>1</td><td>-1</td><td>1</td><td>-1</td><td>$w_3 = t_1 - t_2 + t_3 - t_4$</td></tr><tr><td>$\Gamma^4$</td><td>1</td><td>-1</td><td>-1</td><td>1</td><td>$w_4 = t_1 - t_2 - t_3 + t_4$</td></tr></table>		I	(12)(34)	(13)(24)	(14)(23)	Basis functions	Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$	Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$	Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$	Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$	<table><tr><td></td><td>I</td><td>(12)(34)</td><td>(13)(24)</td><td>(14)(23)</td><td>Basis functions</td></tr><tr><td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>$w_1 = t_1 + t_2 + t_3 + t_4$</td></tr><tr><td>$\Gamma^2$</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>$w_2 = t_1 + t_2 - t_3 - t_4$</td></tr><tr><td>$\Gamma^3$</td><td>1</td><td>-1</td><td>1</td><td>-1</td><td>$w_3 = t_1 - t_2 + t_3 - t_4$</td></tr><tr><td>$\Gamma^4$</td><td>1</td><td>-1</td><td>-1</td><td>1</td><td>$w_4 = t_1 - t_2 - t_3 + t_4$</td></tr></table>		I	(12)(34)	(13)(24)	(14)(23)	Basis functions	Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$	Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$	Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$	Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$
	I	(12)(34)	(13)(24)	(14)(23)	Basis functions																																																												
Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$																																																												
Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$																																																												
Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$																																																												
Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$																																																												
	I	(12)(34)	(13)(24)	(14)(23)	Basis functions																																																												
Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$																																																												
Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$																																																												
Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$																																																												
Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$																																																												
gilmore-lie-groups_ TAB_003	62	0.837	$\begin{tabular}{t}{ l l }\hline & \text{Group \& Dimension} \\ \hline (U\,T(p,\,q)\backslash) & (p^2+q^2+p\,q) \\ (H\,T(p,\,q)\backslash) & (p(p+q)) \\ (U\,T(p,\,q,\,r)\backslash) & (p^2+q^2+r^2+p\,q+p\,r+q\,r) \\ (\operatorname{Sol}(n)\backslash) & (n(n+1)/2) \\ (\operatorname{Nil}(n)\backslash) & (n(n-1)/2) \\ (A(p,\,q)\backslash) & (p\,q) \\ \hline\end{tabular}$	<table><tr><td>Group</td><td>Dimension</td></tr><tr><td>$UT(p, q)$</td><td>$p^2 + q^2 + pq$</td></tr><tr><td>$HT(p, q)$</td><td>$p(p + q)$</td></tr><tr><td>$UT(p, q, r)$</td><td>$p^2 + q^2 + r^2 + pq + pr + qr$</td></tr><tr><td>$Sol(n)$</td><td>$n(n + 1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n - 1)/2$</td></tr><tr><td>$A(p, q)$</td><td>$pq$</td></tr></table>	Group	Dimension	$UT(p, q)$	$p^2 + q^2 + pq$	$HT(p, q)$	$p(p + q)$	$UT(p, q, r)$	$p^2 + q^2 + r^2 + pq + pr + qr$	$Sol(n)$	$n(n + 1)/2$	$Nil(n)$	$n(n - 1)/2$	$A(p, q)$	pq	<table><tr><td>Group</td><td>Dimension</td></tr><tr><td>$UT(p, q)$</td><td>$p^2 + q^2 + pq$</td></tr><tr><td>$HT(p, q)$</td><td>$p(p + q)$</td></tr><tr><td>$UT(p, q, r)$</td><td>$p^2 + q^2 + r^2 + pq + pr + qr$</td></tr><tr><td>$Sol(n)$</td><td>$n(n + 1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n - 1)/2$</td></tr><tr><td>$A(p, q)$</td><td>$pq$</td></tr></table>	Group	Dimension	$UT(p, q)$	$p^2 + q^2 + pq$	$HT(p, q)$	$p(p + q)$	$UT(p, q, r)$	$p^2 + q^2 + r^2 + pq + pr + qr$	$Sol(n)$	$n(n + 1)/2$	$Nil(n)$	$n(n - 1)/2$	$A(p, q)$	pq																																
Group	Dimension																																																																
$UT(p, q)$	$p^2 + q^2 + pq$																																																																
$HT(p, q)$	$p(p + q)$																																																																
$UT(p, q, r)$	$p^2 + q^2 + r^2 + pq + pr + qr$																																																																
$Sol(n)$	$n(n + 1)/2$																																																																
$Nil(n)$	$n(n - 1)/2$																																																																
$A(p, q)$	pq																																																																
Group	Dimension																																																																
$UT(p, q)$	$p^2 + q^2 + pq$																																																																
$HT(p, q)$	$p(p + q)$																																																																
$UT(p, q, r)$	$p^2 + q^2 + r^2 + pq + pr + qr$																																																																
$Sol(n)$	$n(n + 1)/2$																																																																
$Nil(n)$	$n(n - 1)/2$																																																																
$A(p, q)$	pq																																																																
gilmore-lie-groups_ TAB_004	62	1.000	$\begin{tabular}{t}{ l l l }\hline & \text{Transformation \& } (\mathbf{F} = d\mathbf{p}/dt) \\ \hline \text{Galilean} & \text{invariant} & \text{not invariant} \\ \text{Poincaré} & \text{not invariant} & \text{invariant} \\ \hline\end{tabular}$	<table><tr><td>Transformation</td><td>$\mathbf{F} = d\mathbf{p}/dt$</td><td>$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) A_\mu = -\frac{4\pi}{c} j_\mu$</td></tr><tr><td>Galilean</td><td>invariant</td><td>not invariant</td></tr><tr><td>Poincaré</td><td>not invariant</td><td>invariant</td></tr></table>	Transformation	$\mathbf{F} = d\mathbf{p}/dt$	$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) A_\mu = -\frac{4\pi}{c} j_\mu$	Galilean	invariant	not invariant	Poincaré	not invariant	invariant	<table><tr><td>Transformation</td><td>$\mathbf{F} = d\mathbf{p}/dt$</td><td>$(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) A_\mu = -\frac{4\pi}{c} j_\mu$</td></tr><tr><td>Galilean</td><td>invariant</td><td>not invariant</td></tr><tr><td>Poincaré</td><td>not invariant</td><td>invariant</td></tr></table>	Transformation	$\mathbf{F} = d\mathbf{p}/dt$	$(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) A_\mu = -\frac{4\pi}{c} j_\mu$	Galilean	invariant	not invariant	Poincaré	not invariant	invariant																																										
Transformation	$\mathbf{F} = d\mathbf{p}/dt$	$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) A_\mu = -\frac{4\pi}{c} j_\mu$																																																															
Galilean	invariant	not invariant																																																															
Poincaré	not invariant	invariant																																																															
Transformation	$\mathbf{F} = d\mathbf{p}/dt$	$(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) A_\mu = -\frac{4\pi}{c} j_\mu$																																																															
Galilean	invariant	not invariant																																																															
Poincaré	not invariant	invariant																																																															

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																									
gilmore-lie-groups_ TAB_010	167	■ 0.608	$\begin{tabular}{t}{ l l l } \hline \left[H_1, H_2\right] \&= \& 0 \\ \left[\mathbf{H}, E_{\pm 2\mathbf{e}_1}\right] \&= \& ((\pm 2 / \sqrt{12}), 0) E_{\pm 2\mathbf{e}_1} \\ \left[\mathbf{H}, E_{\pm 2\mathbf{e}_2}\right] \&= \& (0, \pm 2 / \sqrt{12}) E_{\pm 2\mathbf{e}_2} \\ \left[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}\right] \&= \& (\pm / \sqrt{12}, \pm / \sqrt{12}) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ \left[E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}\right] \&= \& (2 / \sqrt{12}) H_1 \\ \left[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}\right] \&= \& (2 / \sqrt{12}) H_2 \\ \left[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}\right] \&= \& (1 / \sqrt{12})(\pm H_1 \pm H_2) \\ \left[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}\right] \&= \& *(1 / \sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ \left[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}\right] \&= \& (1 / \sqrt{6}) E_{-2\mathbf{e}_1} \\ \left[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}\right] \&= \& (1 / \sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ \left[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}\right] \&= \& (-1 / \sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ \left[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}\right] \&= \& (-1 / \sqrt{6}) E_{+2\mathbf{e}_1} \\ \left[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}\right] \&= \& (-1 / \sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ \left[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}\right] \&= \& *(1 / \sqrt{6}) E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ \left[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}\right] \&= \& (1 / \sqrt{6}) E_{-2\mathbf{e}_2} \\ \left[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}\right] \&= \& (1 / \sqrt{6}) E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ \left[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}\right] \&= \& (-1 / \sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ \left[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}\right] \&= \& (-1 / \sqrt{6}) E_{+2\mathbf{e}_2} \\ \left[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}\right] \&= \& (-1 / \sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{tabular}$	<table><tr><td>$[H_1, H_2]$</td><td>=</td><td>0</td></tr><tr><td>$[\mathbf{H}, E_{\pm 2\mathbf{e}_1}]$</td><td>=</td><td>$(\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1}$</td></tr><tr><td>$[\mathbf{H}, E_{\pm 2\mathbf{e}_2}]$</td><td>=</td><td>$(0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2}$</td></tr><tr><td>$[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}]$</td><td>=</td><td>$(\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}$</td></tr><tr><td>$[E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}]$</td><td>=</td><td>$(2/\sqrt{12})H_1$</td></tr><tr><td>$[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$</td><td>=</td><td>$(2/\sqrt{12})H_2$</td></tr><tr><td>$[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}]$</td><td>=</td><td>$(1/\sqrt{12})(\pm H_1 \pm H_2)$</td></tr><tr><td>$[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}]$</td><td>=</td><td>$*(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$(1/\sqrt{6})E_{-2\mathbf{e}_1}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}]$</td><td>=</td><td>$(1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+2\mathbf{e}_1}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$*(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$(1/\sqrt{6})E_{-2\mathbf{e}_2}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}]$</td><td>=</td><td>$(1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2}$</td></tr><tr><td>$[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+2\mathbf{e}_2}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$</td></tr></table>	$[H_1, H_2]$	=	0	$[\mathbf{H}, E_{\pm 2\mathbf{e}_1}]$	=	$(\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1}$	$[\mathbf{H}, E_{\pm 2\mathbf{e}_2}]$	=	$(0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2}$	$[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}]$	=	$(\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}$	$[E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}]$	=	$(2/\sqrt{12})H_1$	$[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(2/\sqrt{12})H_2$	$[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}]$	=	$(1/\sqrt{12})(\pm H_1 \pm H_2)$	$[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}]$	=	$*(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{-2\mathbf{e}_1}$	$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}]$	=	$(1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+2\mathbf{e}_1}$	$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	=	$*(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2}$	$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{-2\mathbf{e}_2}$	$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2}$	$[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+2\mathbf{e}_2}$	$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$	
$[H_1, H_2]$	=	0																																																												
$[\mathbf{H}, E_{\pm 2\mathbf{e}_1}]$	=	$(\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1}$																																																												
$[\mathbf{H}, E_{\pm 2\mathbf{e}_2}]$	=	$(0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2}$																																																												
$[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}]$	=	$(\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}$																																																												
$[E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}]$	=	$(2/\sqrt{12})H_1$																																																												
$[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(2/\sqrt{12})H_2$																																																												
$[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}]$	=	$(1/\sqrt{12})(\pm H_1 \pm H_2)$																																																												
$[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}]$	=	$*(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2}$																																																												
$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{-2\mathbf{e}_1}$																																																												
$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}]$	=	$(1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$																																																												
$[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$																																																												
$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+2\mathbf{e}_1}$																																																												
$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$																																																												
$[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	=	$*(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2}$																																																												
$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{-2\mathbf{e}_2}$																																																												
$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2}$																																																												
$[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$																																																												
$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+2\mathbf{e}_2}$																																																												
$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$																																																												

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																																																
gilmore-lie-groups_ TAB_011	172	■1.000	<pre>\begin{tabular}[t]{l l l l l l} \hline \multirow{b}{3}{*}{\((i \ \phi)^{j} \ f_{j}\)} & & & & & \\ \multicolumn{4}{ c }{Representation} \ \hline & & & & & \\ \frac{1}{2} & 1 & & & 2 & \\ & 3 & & & 5 & \\ \hline f_0 & \cos(\phi/2) & 1 & \frac{9}{8} \cos(\frac{\phi}{2}) - \frac{1}{8} \cos(\frac{3\phi}{2}) & 1 & \\ (i \ \phi)^1 f_1 & 2i \sin(\phi/2) & i \sin(\phi) & \frac{9i}{4} \sin(\frac{\phi}{2}) - \frac{i}{12} \sin(\frac{3\phi}{2}) & \frac{i}{3} \sin(\phi) - \frac{i}{6} \sin(2\phi) & \\ (i \ \phi)^2 f_2 & & \cos(\phi) - 1 & -\frac{1}{2} \cos(\frac{\phi}{2}) + \frac{1}{2} \cos(\frac{3\phi}{2}) & -\frac{5}{4} + \frac{1}{3} \cos(\phi) - \frac{1}{12} \cos(2\phi) & \\ (i \ \phi)^3 f_3 & & & -i \sin(\frac{\phi}{2}) + \frac{i}{3} \sin(\frac{3\phi}{2}) & -\frac{i}{3} \sin(\phi) + \frac{i}{6} \sin(2\phi) & \\ (i \ \phi)^4 f_4 & & & & \frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi) & \\ \hline \hline \end{tabular}</pre>	(not rendered)	<table><tr><th colspan="6">Representation</th></tr><tr><th>$(i \phi)^j f_j$</th><th>$\frac{1}{2}$</th><th>1</th><th>$\frac{3}{2}$</th><th>2</th><th></th></tr><tr><td>f_0</td><td>$\cos(\phi/2)$</td><td>1</td><td>$\frac{9}{8} \cos(\frac{\phi}{2}) - \frac{1}{8} \cos(\frac{3\phi}{2})$</td><td>1</td><td></td></tr><tr><td>$(i \phi)^1 f_1$</td><td>$2i \sin(\phi/2)$</td><td>$i \sin(\phi)$</td><td>$\frac{9i}{4} \sin(\frac{\phi}{2}) - \frac{i}{12} \sin(\frac{3\phi}{2})$</td><td>$\frac{i}{3} \sin(\phi) - \frac{i}{6} \sin(2\phi)$</td><td></td></tr><tr><td>$(i \phi)^2 f_2$</td><td></td><td>$\cos(\phi) - 1$</td><td>$-\frac{1}{2} \cos(\frac{\phi}{2}) + \frac{1}{2} \cos(\frac{3\phi}{2})$</td><td>$-\frac{5}{4} + \frac{1}{3} \cos(\phi) - \frac{1}{12} \cos(2\phi)$</td><td></td></tr><tr><td>$(i \phi)^3 f_3$</td><td></td><td></td><td>$-i \sin(\frac{\phi}{2}) + \frac{i}{3} \sin(\frac{3\phi}{2})$</td><td>$-\frac{i}{3} \sin(\phi) + \frac{i}{6} \sin(2\phi)$</td><td></td></tr><tr><td>$(i \phi)^4 f_4$</td><td></td><td></td><td></td><td>$\frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi)$</td><td></td></tr></table>	Representation						$(i \phi)^j f_j$	$\frac{1}{2}$	1	$\frac{3}{2}$	2		f_0	$\cos(\phi/2)$	1	$\frac{9}{8} \cos(\frac{\phi}{2}) - \frac{1}{8} \cos(\frac{3\phi}{2})$	1		$(i \phi)^1 f_1$	$2i \sin(\phi/2)$	$i \sin(\phi)$	$\frac{9i}{4} \sin(\frac{\phi}{2}) - \frac{i}{12} \sin(\frac{3\phi}{2})$	$\frac{i}{3} \sin(\phi) - \frac{i}{6} \sin(2\phi)$		$(i \phi)^2 f_2$		$\cos(\phi) - 1$	$-\frac{1}{2} \cos(\frac{\phi}{2}) + \frac{1}{2} \cos(\frac{3\phi}{2})$	$-\frac{5}{4} + \frac{1}{3} \cos(\phi) - \frac{1}{12} \cos(2\phi)$		$(i \phi)^3 f_3$			$-i \sin(\frac{\phi}{2}) + \frac{i}{3} \sin(\frac{3\phi}{2})$	$-\frac{i}{3} \sin(\phi) + \frac{i}{6} \sin(2\phi)$		$(i \phi)^4 f_4$				$\frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi)$																																							
Representation																																																																																					
$(i \phi)^j f_j$	$\frac{1}{2}$	1	$\frac{3}{2}$	2																																																																																	
f_0	$\cos(\phi/2)$	1	$\frac{9}{8} \cos(\frac{\phi}{2}) - \frac{1}{8} \cos(\frac{3\phi}{2})$	1																																																																																	
$(i \phi)^1 f_1$	$2i \sin(\phi/2)$	$i \sin(\phi)$	$\frac{9i}{4} \sin(\frac{\phi}{2}) - \frac{i}{12} \sin(\frac{3\phi}{2})$	$\frac{i}{3} \sin(\phi) - \frac{i}{6} \sin(2\phi)$																																																																																	
$(i \phi)^2 f_2$		$\cos(\phi) - 1$	$-\frac{1}{2} \cos(\frac{\phi}{2}) + \frac{1}{2} \cos(\frac{3\phi}{2})$	$-\frac{5}{4} + \frac{1}{3} \cos(\phi) - \frac{1}{12} \cos(2\phi)$																																																																																	
$(i \phi)^3 f_3$			$-i \sin(\frac{\phi}{2}) + \frac{i}{3} \sin(\frac{3\phi}{2})$	$-\frac{i}{3} \sin(\phi) + \frac{i}{6} \sin(2\phi)$																																																																																	
$(i \phi)^4 f_4$				$\frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi)$																																																																																	
gilmore-lie-groups_ TAB_012	175	■0.995	<pre>\begin{tabular}[t]{l l l l l l} \hline \cos ^{2}(\alpha, \beta) & \theta(\alpha, \beta) & n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha} & n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta} & \frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n} & \\ \hline 1 & \frac{\pi}{2} \pm \frac{\pi}{2} & \pm 2 & \pm 2 & 1 & \\ 3 & \frac{\pi}{2} \pm \frac{\pi}{3} & \pm 3 & \pm 1 & 3^{-1} & \\ \frac{2}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 1 & \pm 3 & 3^{+1} & \\ \frac{2}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 2 & \pm 1 & 2^{-1} & \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{6} & \pm 1 & \pm 2 & 2^{+1} & \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{6} & \pm 1 & \pm 1 & 1 & \\ 0 & \frac{\pi}{2} & 0 & 0 & - & \\ \hline \hline \end{tabular}</pre>	<table><tr><td>$\cos^2(\alpha, \beta)$</td><td>$\theta(\alpha, \beta)$</td><td>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</td><td>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</td><td>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</td></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>3</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td></td><td></td><td>± 1</td><td>± 3</td><td>3^{+1}</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td></td><td></td><td>± 1</td><td>± 2</td><td>2^{+1}</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>-</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	3	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}			± 1	± 3	3^{+1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}			± 1	± 2	2^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	-	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 1	± 3	3^{+1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 2	2^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																																																																	
1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1																																																																																	
3	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}																																																																																	
		± 1	± 3	3^{+1}																																																																																	
$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}																																																																																	
		± 1	± 2	2^{+1}																																																																																	
$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1																																																																																	
0	$\frac{\pi}{2}$	0	0	-																																																																																	
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																																																																	
1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1																																																																																	
$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}																																																																																	
$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 1	± 3	3^{+1}																																																																																	
$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}																																																																																	
$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 2	2^{+1}																																																																																	
$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1																																																																																	
0	$\frac{\pi}{2}$	0	0	–																																																																																	

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																								
gilmore-lie-groups_ TAB_013	180	0.508	$\begin{tabular}{t}{ l l l l l l }\hline & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ \hline A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_l \\ C_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & 2\mathbf{e}_l \\ \hline \end{tabular}$	<table><tr><td></td><td>α_1</td><td>α_2</td><td>α_3</td><td></td><td>α_{l-1}</td><td>α_l</td></tr><tr><td>A_{l-1}</td><td>$\mathbf{e}_1 - \mathbf{e}_2$</td><td>$\mathbf{e}_2 - \mathbf{e}_3$</td><td>$\mathbf{e}_3 - \mathbf{e}_4$</td><td>...</td><td>$\mathbf{e}_{l-1} - \mathbf{e}_l$</td><td></td></tr><tr><td>D_l</td><td>''</td><td>''</td><td>''</td><td>...</td><td>''</td><td>$\mathbf{e}_{l-1} + \mathbf{e}_l$</td></tr><tr><td>$B_l$</td><td>''</td><td>''</td><td>''</td><td>...</td><td>''</td><td>$\mathbf{e}_l$</td></tr><tr><td>C_l</td><td>''</td><td>''</td><td>''</td><td>...</td><td>''</td><td>$2\mathbf{e}_l$</td></tr></table>		α_1	α_2	α_3		α_{l-1}	α_l	A_{l-1}	$\mathbf{e}_1 - \mathbf{e}_2$	$\mathbf{e}_2 - \mathbf{e}_3$	$\mathbf{e}_3 - \mathbf{e}_4$	...	$\mathbf{e}_{l-1} - \mathbf{e}_l$		D_l	''	''	''	...	''	$\mathbf{e}_{l-1} + \mathbf{e}_l$	B_l	''	''	''	...	''	$\mathbf{e}_l$	C_l	''	''	''	...	''	$2\mathbf{e}_l$	$\begin{array}{ccccccc} & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_l \\ C_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & 2\mathbf{e}_l \end{array}$					
	α_1	α_2	α_3		α_{l-1}	α_l																																							
A_{l-1}	$\mathbf{e}_1 - \mathbf{e}_2$	$\mathbf{e}_2 - \mathbf{e}_3$	$\mathbf{e}_3 - \mathbf{e}_4$	...	$\mathbf{e}_{l-1} - \mathbf{e}_l$																																								
D_l	''	''	''	...	''	$\mathbf{e}_{l-1} + \mathbf{e}_l$																																							
B_l	''	''	''	...	''	$\mathbf{e}_l$																																							
C_l	''	''	''	...	''	$2\mathbf{e}_l$																																							
gilmore-lie-groups_ TAB_014	182	0.999	$\begin{tabular}{t}{ l l l l }\hline & \alpha_1 & \alpha_2 & \alpha_3 \\ \hline A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 \\ D_l & \text{''} & \text{''} & \text{''} \\ B_l & \text{''} & \text{''} & \text{''} \\ C_l & \text{''} & \text{''} & \text{''} \\ \hline \end{tabular}$	<table><tr><td>p</td><td>q</td><td>r</td><td>Root space</td></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8	<table><tr><th>p</th><th>q</th><th>r</th><th>Root space</th></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8
p	q	r	Root space																																										
p	2	2	D_{p+2}																																										
3	3	2	E_6																																										
4	3	2	E_7																																										
5	3	2	E_8																																										
p	q	r	Root space																																										
p	2	2	D_{p+2}																																										
3	3	2	E_6																																										
4	3	2	E_7																																										
5	3	2	E_8																																										
gilmore-lie-groups_ TAB_015	184	1.000	$\begin{tabular}{t}{ l l l l }\hline & \alpha_1 & \alpha_2 & \alpha_3 \\ \hline A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 \\ D_l & \text{''} & \text{''} & \text{''} \\ B_l & \text{''} & \text{''} & \text{''} \\ C_l & \text{''} & \text{''} & \text{''} \\ \hline \end{tabular}$	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i$</td></tr><tr><td>$E_{+\mathbf{e}_i - \mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$	$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i$</td></tr><tr><td>$E_{+\mathbf{e}_i - \mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$	$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$																
Canonical form	Boson operators	Coordinates and derivatives	Fermion operators																																										
H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$																																										
$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$																																										
Canonical form	Boson operators	Coordinates and derivatives	Fermion operators																																										
H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$																																										
$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$																																										

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																																																										
gilmore-lie-groups_ TAB_018	195	■ 0.500	$\begin{tabular}{t l l l l l } \hline \scriptstyle (A_1) & \sim & \scriptstyle (B_1) & \sim & \scriptstyle (C_1) \\ \scriptstyle (B_1) & \sim & \scriptstyle (C_1) & \sim & \scriptstyle (C_1) \\ \hline \scriptstyle (u(2)) & \sim & \scriptstyle (so(3)) & \sim & \scriptstyle (sp(1) = usp(2)) \\ \scriptstyle (u(1,1) = sl(2; \mathbb{R})) & \sim & \scriptstyle (so(2,1)) & \sim & \scriptstyle (sp(2; \mathbb{R})) \\ \hline \scriptstyle (D_2) & = & \scriptstyle (A_1) & + & \scriptstyle (A_1) \\ \scriptstyle (so(4)) & = & \scriptstyle (so(3)) & + & \scriptstyle (so(3)) \\ \scriptstyle (so^*(4)) & \sim & \scriptstyle (so(3)) & + & \scriptstyle (so(2,1)) \\ \scriptstyle (so(3,1)) & \sim & \scriptstyle (sl(2; \mathbb{C})) & & \\ \scriptstyle (so(2,2)) & \sim & \scriptstyle (so(2,1)) & + & \scriptstyle (so(2,1)) \\ \hline \scriptstyle (B_2) & = & \scriptstyle (C_2) & & \\ \scriptstyle (so(5)) & \sim & \scriptstyle (sp(2) = usp(4)) & & \\ \scriptstyle (so(4,1)) & \sim & \scriptstyle (sp(1,1) = usp(2,2)) & & \\ \scriptstyle (so(3,2)) & \sim & \scriptstyle (sp(4; R)) & & \\ \hline \scriptstyle (D_3) & = & \scriptstyle (A_3) & & \\ \scriptstyle (so(6)) & \sim & \scriptstyle (su(4)) & & \\ \scriptstyle (so(5,1)) & \sim & \scriptstyle (su^*(4)) & & \\ \scriptstyle (so^*(6)) & \sim & \scriptstyle (su(3,1)) & & \\ \scriptstyle (so(4,2)) & \sim & \scriptstyle (su(2,2)) & & \\ \scriptstyle (so(3,3)) & \sim & \scriptstyle (sl(4; \mathbb{R})) & & \\ \hline \end{tabular}$	<table><tr><td>A_1</td><td></td><td>B_1</td><td></td><td>C_1</td></tr><tr><td>$su(2)$</td><td></td><td>$so(3)$</td><td></td><td>$sp(1) = usp(2)$</td></tr><tr><td>$su(1, 1) = sl(2; \mathbb{R})$</td><td></td><td>$so(2, 1)$</td><td></td><td>$sp(2; \mathbb{R})$</td></tr><tr><td>$D_2$</td><td>$=$</td><td>$A_1$</td><td>$+$</td><td>$A_1$</td></tr><tr><td>$so(4)$</td><td>$=$</td><td>$so(3)$</td><td>$+$</td><td>$so(3)$</td></tr><tr><td>$so^*(4)$</td><td></td><td>$so(3)$</td><td>$+$</td><td>$so(2, 1)$</td></tr><tr><td>$so(3, 1)$</td><td></td><td>$sl(2; \mathbb{C})$</td><td></td><td></td></tr><tr><td>$so(2, 2)$</td><td></td><td>$so(2, 1)$</td><td>$+$</td><td>$so(2, 1)$</td></tr><tr><td>B_2</td><td>$=$</td><td>C_2</td><td></td><td></td></tr><tr><td>$so(5)$</td><td>$\sim$</td><td>$sp(2) = usp(4)$</td><td></td><td></td></tr><tr><td>$so(4, 1)$</td><td>$\sim$</td><td>$sp(1, 1) = usp(2, 2)$</td><td></td><td></td></tr><tr><td>$so(3, 2)$</td><td>$\sim$</td><td>$sp(4; R)$</td><td></td><td></td></tr><tr><td>D_3</td><td>$=$</td><td>A_3</td><td></td><td></td></tr><tr><td>$so(6)$</td><td>$\sim$</td><td>$su(4)$</td><td></td><td></td></tr><tr><td>$so(5, 1)$</td><td>$\sim$</td><td>$su^*(4)$</td><td></td><td></td></tr><tr><td>$so^*(6)$</td><td>$\sim$</td><td>$su(3, 1)$</td><td></td><td></td></tr><tr><td>$so(4, 2)$</td><td>$\sim$</td><td>$su(2, 2)$</td><td></td><td></td></tr><tr><td>$so(3, 3)$</td><td>$\sim$</td><td>$sl(4; \mathbb{R})$</td><td></td><td></td></tr></table>	A_1		B_1		C_1	$su(2)$		$so(3)$		$sp(1) = usp(2)$	$su(1, 1) = sl(2; \mathbb{R})$		$so(2, 1)$		$sp(2; \mathbb{R})$	D_2	$=$	A_1	$+$	A_1	$so(4)$	$=$	$so(3)$	$+$	$so(3)$	$so^*(4)$		$so(3)$	$+$	$so(2, 1)$	$so(3, 1)$		$sl(2; \mathbb{C})$			$so(2, 2)$		$so(2, 1)$	$+$	$so(2, 1)$	B_2	$=$	C_2			$so(5)$	$\sim$	$sp(2) = usp(4)$			$so(4, 1)$	$\sim$	$sp(1, 1) = usp(2, 2)$			$so(3, 2)$	$\sim$	$sp(4; R)$			D_3	$=$	A_3			$so(6)$	$\sim$	$su(4)$			$so(5, 1)$	$\sim$	$su^*(4)$			$so^*(6)$	$\sim$	$su(3, 1)$			$so(4, 2)$	$\sim$	$su(2, 2)$			$so(3, 3)$	$\sim$	$sl(4; \mathbb{R})$			
A_1		B_1		C_1																																																																																											
$su(2)$		$so(3)$		$sp(1) = usp(2)$																																																																																											
$su(1, 1) = sl(2; \mathbb{R})$		$so(2, 1)$		$sp(2; \mathbb{R})$																																																																																											
D_2	$=$	A_1	$+$	A_1																																																																																											
$so(4)$	$=$	$so(3)$	$+$	$so(3)$																																																																																											
$so^*(4)$		$so(3)$	$+$	$so(2, 1)$																																																																																											
$so(3, 1)$		$sl(2; \mathbb{C})$																																																																																													
$so(2, 2)$		$so(2, 1)$	$+$	$so(2, 1)$																																																																																											
B_2	$=$	C_2																																																																																													
$so(5)$	$\sim$	$sp(2) = usp(4)$																																																																																													
$so(4, 1)$	$\sim$	$sp(1, 1) = usp(2, 2)$																																																																																													
$so(3, 2)$	$\sim$	$sp(4; R)$																																																																																													
D_3	$=$	A_3																																																																																													
$so(6)$	$\sim$	$su(4)$																																																																																													
$so(5, 1)$	$\sim$	$su^*(4)$																																																																																													
$so^*(6)$	$\sim$	$su(3, 1)$																																																																																													
$so(4, 2)$	$\sim$	$su(2, 2)$																																																																																													
$so(3, 3)$	$\sim$	$sl(4; \mathbb{R})$																																																																																													

Identifier	Page	Conf.	LaTeX source	Rendered	Image																														
gilmore-lie-groups_ TAB_019	196	■1.000	<pre>\begin{tabular}[t]{ l l l l } \hline \multirow[b]{2}{*}{Root space} & \multirow[b]{2}{*}{Class \({}_\text{rank(character)}\)} & \multicolumn{2}{ l }{Maximal compact subgroup} \\ \hline & & Root space & Dimension \\ \hline G_2 & G_{2(-14)} & G_2 & 14 \\ & G_{2(+2)} & A_1 + A_1 & 6 \\ F_4 & F_{4(-52)} & F_4 & 52 \\ & F_{4(-20)} & B_4 & 36 \\ & F_{4(+4)} & C_3 + A_1 & 24 \\ E_6 & E_{6(-78)} & E_6 & 78 \\ & E_{6(-26)} & F_4 & 52 \\ & E_{6(-14)} & D_5 + D_1 & 46 \\ & E_{6(+2)} & A_5 + A_1 & 38 \\ & E_{6(+6)} & C_4 & 36 \\ E_7 & E_{7(-133)} & E_7 & 133 \\ & E_{7(-25)} & E_6 + D_1 & 79 \\ & E_{7(-5)} & D_6 + A_1 & 69 \\ & E_{7(+7)} & A_7 & 63 \\ E_8 & E_{8(-248)} & E_8 & 248 \\ & E_{8(-24)} & E_7 + A_1 & 136 \\ & E_{8(+8)} & D_8 & 120 \\ \hline \hline \end{tabular}</pre>	(not rendered)	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$															
Quadrant	Operator	Edge																																	
I	J_+	$n_2 = 0$																																	
I	J_-	$n_1 = 0$																																	
III	J_+	$n_1 = -1$																																	
III	J_-	$n_2 = -1$																																	
gilmore-lie-groups_ TAB_020	199	■1.000	<pre>\begin{tabular}[t]{ l l l l } \hline Quadrant & Operator & Edge \\ \hline I & \({}_J\{+\}\) & \({}_n\{2\}=0\} \\ \hline I & \({}_J\{-\}\) & \({}_n\{1\}=0\} \\ \hline III & \({}_J\{+\}\) & \({}_n\{1\}=-1\} \\ \hline III & \({}_J\{-\}\) & \({}_n\{2\}=-1\} \\ \hline \end{tabular}</pre>	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$
Quadrant	Operator	Edge																																	
I	J_+	$n_2 = 0$																																	
I	J_-	$n_1 = 0$																																	
III	J_+	$n_1 = -1$																																	
III	J_-	$n_2 = -1$																																	
Quadrant	Operator	Edge																																	
I	J_+	$n_2 = 0$																																	
I	J_-	$n_1 = 0$																																	
III	J_+	$n_1 = -1$																																	
III	J_-	$n_2 = -1$																																	

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																																														
gilmore-lie-groups_ TAB_021	206	0.877	$\begin{tabular}{t l l l l l } \hline \text{Root space} \& \text{Quotient } (G^{\prime}/H) \& \text{Dimension } (P) \& \text{Rank } (P) \\ \hline (A_{p+q-1}) \& (SU(p, q)/S[U(p) \otimes U(q)]) \& 2pq \& \min(p, q) \\ (A_{n-1}) \& (SL(n; \mathbb{R})/SO(n)) \& \frac{1}{2}(n+2)(n-1) \& n-1 \\ (A_{2n-1}) \& (SU^*(2n)/USp(2n)) \& (2n+1)(n-1) \& n-1 \\ (B_{p+q}) \& (SO(p, q)/SO(p) \otimes SO(q)) \& pq \& \min(p, q) \\ (D_{p+q}) \& (SO(p, q)/SO(p) \otimes SO(q)) \& pq \& \min(p, q) \\ (D_n) \& (SO^*(2n)/U(n)) \& n(n-1) \& [n/2] \\ (C_{p+q}) \& (USp(2p, 2q)/USp(2p) \otimes USp(2q)) \& 4pq \& \min(p, q) \\ (C_n) \& (Sp(2n; \mathbb{R})/U(n)) \& n(n+1) \& n \\ \hline \end{tabular}$	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S[U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S[U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S[U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S[U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n						
Root space	Quotient G'/H	Dimension P	Rank P																																																																																
A_{p+q-1}	$SU(p, q)/S[U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$																																																																																
A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$																																																																																
A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$																																																																																
B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																																																																																
D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																																																																																
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																																																																																
C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$																																																																																
C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n																																																																																
Root space	Quotient G'/H	Dimension P	Rank P																																																																																
A_{p+q-1}	$SU(p, q)/S[U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$																																																																																
A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$																																																																																
A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$																																																																																
B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																																																																																
D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																																																																																
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																																																																																
C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$																																																																																
C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n																																																																																
gilmore-lie-groups_ TAB_022	207	0.979	$\begin{tabular}{t l l l l l l } \hline \text{Root space} \& (G^{\prime}/H) \& (\operatorname{Dim} G^{\prime}) \& (\operatorname{Dim} H) \& (\operatorname{Dim} P) \& \text{Rank } (P) \\ \hline (G_2) \& (G_{2(+2)})/(A_1 \oplus A_1) \& 14 \& 6 \& 8 \& 2 \\ (F_4) \& (F_{4(-20)}/B_4) \& 52 \& 36 \& 16 \& 1 \\ (F_4) \& (F_{4(+4)}/(C_3 \oplus A_1)) \& 52 \& 24 \& 28 \& 4 \\ (E_6) \& (E_{6(-26)}/F_4) \& 78 \& 52 \& 26 \& 2 \\ (E_6) \& (E_{6(-14)}/(D_5 \oplus D_1)) \& 78 \& 46 \& 32 \& 2 \\ (E_6) \& (E_{6(+2)}/(A_5 \oplus A_1)) \& 78 \& 38 \& 40 \& 4 \\ (E_6) \& (E_{6(+6)}/C_4) \& 78 \& 36 \& 42 \& 6 \\ (E_7) \& (E_{7(-25)}/(E_6 \oplus D_1)) \& 133 \& 79 \& 54 \& 3 \\ (E_7) \& (E_{7(-5)}/(D_6 \oplus A_1)) \& 133 \& 69 \& 64 \& 4 \\ (E_7) \& (E_{7(+7)}/A_7) \& 133 \& 63 \& 70 \& 7 \\ (E_8) \& (E_{8(-24)}/(E_7 \oplus A_1)) \& 248 \& 136 \& 112 \& 4 \\ (E_8) \& (E_{8(+8)}/D_8) \& 248 \& 120 \& 128 \& 8 \\ \hline \end{tabular}$	(not rendered)	<table><tr><th>Root space</th><th>G'/H</th><th>Dim G'</th><th>Dim H</th><th>Dim P</th><th>Rank P</th></tr><tr><td>G_2</td><td>$G_{2(+2)}/(A_1 \oplus A_1)$</td><td>14</td><td>6</td><td>8</td><td>2</td></tr><tr><td>F_4</td><td>$F_{4(-20)}/B_4$</td><td>52</td><td>36</td><td>16</td><td>1</td></tr><tr><td></td><td>$F_{4(+4)}/(C_3 \oplus A_1)$</td><td>52</td><td>24</td><td>28</td><td>4</td></tr><tr><td>E_6</td><td>$E_{6(-26)}/F_4$</td><td>78</td><td>52</td><td>26</td><td>2</td></tr><tr><td></td><td>$E_{6(-14)}/(D_5 \oplus D_1)$</td><td>78</td><td>46</td><td>32</td><td>2</td></tr><tr><td></td><td>$E_{6(+2)}/(A_5 \oplus A_1)$</td><td>78</td><td>38</td><td>40</td><td>4</td></tr><tr><td></td><td>$E_{6(+6)}/C_4$</td><td>78</td><td>36</td><td>42</td><td>6</td></tr><tr><td>E_7</td><td>$E_{7(-25)}/(E_6 \oplus D_1)$</td><td>133</td><td>79</td><td>54</td><td>3</td></tr><tr><td></td><td>$E_{7(-5)}/(D_6 \oplus A_1)$</td><td>133</td><td>69</td><td>64</td><td>4</td></tr><tr><td></td><td>$E_{7(+7)}/A_7$</td><td>133</td><td>63</td><td>70</td><td>7</td></tr><tr><td>E_8</td><td>$E_{8(-24)}/(E_7 \oplus A_1)$</td><td>248</td><td>136</td><td>112</td><td>4</td></tr><tr><td></td><td>$E_{8(+8)}/D_8$</td><td>248</td><td>120</td><td>128</td><td>8</td></tr></table>	Root space	G'/H	Dim G'	Dim H	Dim P	Rank P	G_2	$G_{2(+2)}/(A_1 \oplus A_1)$	14	6	8	2	F_4	$F_{4(-20)}/B_4$	52	36	16	1		$F_{4(+4)}/(C_3 \oplus A_1)$	52	24	28	4	E_6	$E_{6(-26)}/F_4$	78	52	26	2		$E_{6(-14)}/(D_5 \oplus D_1)$	78	46	32	2		$E_{6(+2)}/(A_5 \oplus A_1)$	78	38	40	4		$E_{6(+6)}/C_4$	78	36	42	6	E_7	$E_{7(-25)}/(E_6 \oplus D_1)$	133	79	54	3		$E_{7(-5)}/(D_6 \oplus A_1)$	133	69	64	4		$E_{7(+7)}/A_7$	133	63	70	7	E_8	$E_{8(-24)}/(E_7 \oplus A_1)$	248	136	112	4		$E_{8(+8)}/D_8$	248	120	128	8
Root space	G'/H	Dim G'	Dim H	Dim P	Rank P																																																																														
G_2	$G_{2(+2)}/(A_1 \oplus A_1)$	14	6	8	2																																																																														
F_4	$F_{4(-20)}/B_4$	52	36	16	1																																																																														
	$F_{4(+4)}/(C_3 \oplus A_1)$	52	24	28	4																																																																														
E_6	$E_{6(-26)}/F_4$	78	52	26	2																																																																														
	$E_{6(-14)}/(D_5 \oplus D_1)$	78	46	32	2																																																																														
	$E_{6(+2)}/(A_5 \oplus A_1)$	78	38	40	4																																																																														
	$E_{6(+6)}/C_4$	78	36	42	6																																																																														
E_7	$E_{7(-25)}/(E_6 \oplus D_1)$	133	79	54	3																																																																														
	$E_{7(-5)}/(D_6 \oplus A_1)$	133	69	64	4																																																																														
	$E_{7(+7)}/A_7$	133	63	70	7																																																																														
E_8	$E_{8(-24)}/(E_7 \oplus A_1)$	248	136	112	4																																																																														
	$E_{8(+8)}/D_8$	248	120	128	8																																																																														

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																
gilmore-lie-groups_ TAB_023	212	■ 1.000	$\begin{tabular}{t}{ l l l l } \hline \text{Operation} \& \backslash(i \sigma_1 \backslash) \& \backslash(i \sigma_2 \backslash) \& \backslash(i \sigma_3 \backslash) \\ \hline \backslash(T_1=\backslash) \text{block matrix decomposition} \& -1 \& -1 \& +1 \\ \hline \backslash(T_2=\backslash) \text{complex conjugation} \& -1 \& +1 \& -1 \\ \hline \backslash(T_3=T_1 T_2) \& +1 \& -1 \& -1 \\ \hline \end{tabular}$	<table><tr><th>Operation</th><th>$i\sigma_1$</th><th>$i\sigma_2$</th><th>$i\sigma_3$</th></tr><tr><td>$T_1 = \text{block matrix decomposition}$</td><td>-1</td><td>-1</td><td>+1</td></tr><tr><td>$T_2 = \text{complex conjugation}$</td><td>-1</td><td>+1</td><td>-1</td></tr><tr><td>$T_3 = T_1 T_2$</td><td>+1</td><td>-1</td><td>-1</td></tr></table>	Operation	$i\sigma_1$	$i\sigma_2$	$i\sigma_3$	$T_1 = \text{block matrix decomposition}$	-1	-1	+1	$T_2 = \text{complex conjugation}$	-1	+1	-1	$T_3 = T_1 T_2$	+1	-1	-1	<table><tr><th>Operation</th><th>$i\sigma_1$</th><th>$i\sigma_2$</th><th>$i\sigma_3$</th></tr><tr><td>$T_1 = \text{block matrix decomposition}$</td><td>-1</td><td>-1</td><td>+1</td></tr><tr><td>$T_2 = \text{complex conjugation}$</td><td>-1</td><td>+1</td><td>-1</td></tr><tr><td>$T_3 = T_1 T_2$</td><td>+1</td><td>-1</td><td>-1</td></tr></table>	Operation	$i\sigma_1$	$i\sigma_2$	$i\sigma_3$	$T_1 = \text{block matrix decomposition}$	-1	-1	+1	$T_2 = \text{complex conjugation}$	-1	+1	-1	$T_3 = T_1 T_2$	+1	-1	-1
Operation	$i\sigma_1$	$i\sigma_2$	$i\sigma_3$																																		
$T_1 = \text{block matrix decomposition}$	-1	-1	+1																																		
$T_2 = \text{complex conjugation}$	-1	+1	-1																																		
$T_3 = T_1 T_2$	+1	-1	-1																																		
Operation	$i\sigma_1$	$i\sigma_2$	$i\sigma_3$																																		
$T_1 = \text{block matrix decomposition}$	-1	-1	+1																																		
$T_2 = \text{complex conjugation}$	-1	+1	-1																																		
$T_3 = T_1 T_2$	+1	-1	-1																																		
gilmore-lie-groups_ TAB_024	232	■ 1.000	$\begin{tabular}{t}{ l l l l } \hline \text{Bessel function} \& \rightarrow \& \text{harmonic oscillator eigenfunction} \\ \hline \text{Bessel equation} \& \rightarrow \& \text{Schrödinger equation for harmonic oscillator} \\ \hline \end{tabular}$	<table><tr><td>Bessel function</td><td>→</td><td>harmonic oscillator eigenfunction</td></tr><tr><td>Bessel equation</td><td>→</td><td>Schrödinger equation for harmonic oscillator</td></tr></table>	Bessel function	→	harmonic oscillator eigenfunction	Bessel equation	→	Schrödinger equation for harmonic oscillator	<table><tr><td>Bessel function</td><td>→</td><td>harmonic oscillator eigenfunction</td></tr><tr><td>Bessel equation</td><td>→</td><td>Schrödinger equation for harmonic oscillator</td></tr></table>	Bessel function	→	harmonic oscillator eigenfunction	Bessel equation	→	Schrödinger equation for harmonic oscillator																				
Bessel function	→	harmonic oscillator eigenfunction																																			
Bessel equation	→	Schrödinger equation for harmonic oscillator																																			
Bessel function	→	harmonic oscillator eigenfunction																																			
Bessel equation	→	Schrödinger equation for harmonic oscillator																																			
gilmore-lie-groups_ TAB_025	239	■ 0.424	$\begin{tabular}{t}{ l l l l l } \hline \text{Equation} \& \backslash(A \backslash) \& \backslash(B \backslash) \& \backslash(C \backslash) \\ \hline \text{Klein-Gordon} \& \backslash(-l(l+1)+\left(e^2 / \hbar c\right)^2 \backslash) \& \backslash(2 E e^2 /(\hbar c) \backslash) \& \backslash(\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2 \backslash) \\ \hline \text{Schrödinger} \& \backslash(-l(l+1) \backslash) \& \backslash(2 m e^2 / \hbar^2 \backslash) \& \backslash(2 m \mathrm{~} W / \hbar^2 \backslash) \\ \hline \end{tabular}$	<table><tr><th>Equation</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Klein-Gordon</td><td>$-l(l+1)+(e^2/\hbar c)^2$</td><td>$2Ee^2/(\hbar c)^2$</td><td>$[E^2-(mc^2)^2]/(\hbar c)^2$</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2me^2/\hbar^2$</td><td>$2mW/\hbar^2$</td></tr></table>	Equation	A	B	C	Klein-Gordon	$-l(l+1)+(e^2/\hbar c)^2$	$2Ee^2/(\hbar c)^2$	$[E^2-(mc^2)^2]/(\hbar c)^2$	Schrödinger	$-l(l+1)$	$2me^2/\hbar^2$	$2mW/\hbar^2$	<table><tr><th>Equation</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Klein–Gordon</td><td>$-l(l+1)+(e^2/\hbar c)^2$</td><td>$2Ee^2/(\hbar c)^2$</td><td>$[E^2-(mc^2)^2]/(\hbar c)^2$</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2me^2/\hbar^2$</td><td>$2mW/\hbar^2$</td></tr></table>	Equation	A	B	C	Klein–Gordon	$-l(l+1)+(e^2/\hbar c)^2$	$2Ee^2/(\hbar c)^2$	$[E^2-(mc^2)^2]/(\hbar c)^2$	Schrödinger	$-l(l+1)$	$2me^2/\hbar^2$	$2mW/\hbar^2$								
Equation	A	B	C																																		
Klein-Gordon	$-l(l+1)+(e^2/\hbar c)^2$	$2Ee^2/(\hbar c)^2$	$[E^2-(mc^2)^2]/(\hbar c)^2$																																		
Schrödinger	$-l(l+1)$	$2me^2/\hbar^2$	$2mW/\hbar^2$																																		
Equation	A	B	C																																		
Klein–Gordon	$-l(l+1)+(e^2/\hbar c)^2$	$2Ee^2/(\hbar c)^2$	$[E^2-(mc^2)^2]/(\hbar c)^2$																																		
Schrödinger	$-l(l+1)$	$2me^2/\hbar^2$	$2mW/\hbar^2$																																		

Identifier	Page	Conf.	LaTeX source	Rendered	Image																				
gilmore-lie-groups_ TAB_026	240	■ 0.655	<pre>\begin{tabular}[t]{l l l l}\hline & Procedure & Result \\ \hline 1 & Locate singularities & $0, \infty$ \\ 2 & Determine analytic behavior at singular points & $r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$ \\ 3 & Keep only $\mathcal{L}^2$ solutions & $\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$ \\ 4 & Look for solutions with proper asymptotic behavior & $R = r^\gamma e^{\lambda r} f(r)$ \\ 5 & Construct differential equation for $f(r)$ & $[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$ \\ 6 & Construct recursion relation & $f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$ \\ 7 & Look at asymptotic behavior & $f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$) \\ 8 & Construct quantization condition & $2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$ \\ 9 & Construct explicit solutions & $E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$ \\ \hline \end{tabular}</pre>	(not rendered)	<table><tr><th>Procedure</th><th>Result</th></tr><tr><td>1 Locate singularities</td><td>$0, \infty$</td></tr><tr><td>2 Determine analytic behavior at singular points</td><td>$r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$</td></tr><tr><td>3 Keep only $\mathcal{L}^2$ solutions</td><td>$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$</td></tr><tr><td>4 Look for solutions with proper asymptotic behavior</td><td>$R = r^\gamma e^{\lambda r} f(r)$</td></tr><tr><td>5 Construct differential equation for $f(r)$</td><td>$[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$</td></tr><tr><td>6 Construct recursion relation</td><td>$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$</td></tr><tr><td>7 Look at asymptotic behavior</td><td>$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)</td></tr><tr><td>8 Construct quantization condition</td><td>$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$</td></tr><tr><td>9 Construct explicit solutions</td><td>$E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$</td></tr></table>	Procedure	Result	1 Locate singularities	$0, \infty$	2 Determine analytic behavior at singular points	$r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$	3 Keep only $\mathcal{L}^2$ solutions	$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$	4 Look for solutions with proper asymptotic behavior	$R = r^\gamma e^{\lambda r} f(r)$	5 Construct differential equation for $f(r)$	$[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$	6 Construct recursion relation	$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$	7 Look at asymptotic behavior	$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)	8 Construct quantization condition	$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$	9 Construct explicit solutions	$E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$
Procedure	Result																								
1 Locate singularities	$0, \infty$																								
2 Determine analytic behavior at singular points	$r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$																								
3 Keep only $\mathcal{L}^2$ solutions	$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$																								
4 Look for solutions with proper asymptotic behavior	$R = r^\gamma e^{\lambda r} f(r)$																								
5 Construct differential equation for $f(r)$	$[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$																								
6 Construct recursion relation	$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$																								
7 Look at asymptotic behavior	$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)																								
8 Construct quantization condition	$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$																								
9 Construct explicit solutions	$E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$																								

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																						
gilmore-lie-groups_TAB_027	244	■0.998	$\begin{tabular}{t l l l l } \hline \backslash & \backslash (l=0) & \backslash (l=1) & \backslash (l=2) & \backslash (l=3) \\ \hline 0 & \sqrt{\frac{1}{4\pi}} & \sqrt{\frac{3}{4\pi}}\cos\theta & \sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1) & \sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta) \\ \pm 1 & & \mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi} & \mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi} & \mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi} \\ \pm 2 & & & \sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi} & \sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi} \\ \pm 3 & & & & \mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi} \\ \hline \end{tabular}$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$				
m	$l=0$	$l=1$	$l=2$	$l=3$																																																							
0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$																																																							
± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$																																																							
± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$																																																							
± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$																																																							
m	$l=0$	$l=1$	$l=2$	$l=3$																																																							
0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$																																																							
± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$																																																							
± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$																																																							
± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$																																																							
gilmore-lie-groups_TAB_028	258	■0.721	$\begin{tabular}{t l l l } \hline H & \backslash (\frac{p^2}{2m}-\frac{\alpha}{r}) & \backslash (\sqrt{p^2+m^2}-\frac{\alpha}{r}) & \backslash (\mathbf{r}\cdot\mathbf{p}-i) \\ \hline A_4 & \backslash (\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)) & \backslash (\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)) & \backslash (\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r})) \\ \hline \Gamma_4 & \backslash (\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)) & \backslash (\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)) & \backslash (\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r})) \\ \hline \Gamma_5 & \backslash (\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)) & \backslash (\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r})) & \backslash (\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r})) \\ \hline \Theta & \backslash (H_S-W) & \backslash (A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C) & \backslash (A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C) \\ \hline A & \backslash (1/2m) & \backslash (m^2-E^2) & \backslash (1) \\ \hline B & \backslash (-W) & \backslash (-2\alpha E) & \backslash (m^2-E^2) \\ \hline C & \backslash (-\alpha) & \backslash (-2\alpha E) & \backslash (-2\alpha E) \\ \hline \end{tabular}$	<table><tr><th>H</th><th>$\frac{p^2}{2m}-\frac{\alpha}{r}$</th><th>$\sqrt{p^2+m^2}-\frac{\alpha}{r}$</th></tr><tr><td>$A_4$</td><td>$\mathbf{r}\cdot\mathbf{p}-i$</td><td>$\mathbf{r}\cdot\mathbf{p}-i$</td></tr><tr><td>$\Gamma_4$</td><td>$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$</td><td>$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Gamma_5$</td><td>$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$</td><td>$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Theta$</td><td>$r(H_S-W)$</td><td>$r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$</td></tr><tr><td></td><td>$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$</td><td>$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$</td></tr><tr><td>$A$</td><td>$1/2m$</td><td>$1$</td></tr><tr><td>$B$</td><td>$-W$</td><td>$m^2-E^2$</td></tr><tr><td>$C$</td><td>$-\alpha$</td><td>$-2\alpha E$</td></tr></table>	H	$\frac{p^2}{2m}-\frac{\alpha}{r}$	$\sqrt{p^2+m^2}-\frac{\alpha}{r}$	A_4	$\mathbf{r}\cdot\mathbf{p}-i$	$\mathbf{r}\cdot\mathbf{p}-i$	Γ_4	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$	Γ_5	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$	Θ	$r(H_S-W)$	$r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$		$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	A	$1/2m$	1	B	$-W$	m^2-E^2	C	$-\alpha$	$-2\alpha E$	<table><tr><th>H</th><th>$\frac{p^2}{2m}-\frac{\alpha}{r}$</th><th>$\sqrt{p^2+m^2}-\frac{\alpha}{r}$</th></tr><tr><td>$A_4$</td><td>$\mathbf{r}\cdot\mathbf{p}-i$</td><td>$\mathbf{r}\cdot\mathbf{p}-i$</td></tr><tr><td>$\Gamma_4$</td><td>$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$</td><td>$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Gamma_5$</td><td>$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$</td><td>$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Theta$</td><td>$r(H_S-W)$</td><td>$r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$</td></tr><tr><td></td><td>$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$</td><td>$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$</td></tr><tr><td>$A$</td><td>$1/2m$</td><td>$1$</td></tr><tr><td>$B$</td><td>$-W$</td><td>$m^2-E^2$</td></tr><tr><td>$C$</td><td>$-\alpha$</td><td>$-2\alpha E$</td></tr></table>	H	$\frac{p^2}{2m}-\frac{\alpha}{r}$	$\sqrt{p^2+m^2}-\frac{\alpha}{r}$	A_4	$\mathbf{r}\cdot\mathbf{p}-i$	$\mathbf{r}\cdot\mathbf{p}-i$	Γ_4	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$	Γ_5	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$	Θ	$r(H_S-W)$	$r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$		$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	A	$1/2m$	1	B	$-W$	m^2-E^2	C	$-\alpha$	$-2\alpha E$
H	$\frac{p^2}{2m}-\frac{\alpha}{r}$	$\sqrt{p^2+m^2}-\frac{\alpha}{r}$																																																									
A_4	$\mathbf{r}\cdot\mathbf{p}-i$	$\mathbf{r}\cdot\mathbf{p}-i$																																																									
Γ_4	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$																																																									
Γ_5	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$																																																									
Θ	$r(H_S-W)$	$r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$																																																									
	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$																																																									
A	$1/2m$	1																																																									
B	$-W$	m^2-E^2																																																									
C	$-\alpha$	$-2\alpha E$																																																									
H	$\frac{p^2}{2m}-\frac{\alpha}{r}$	$\sqrt{p^2+m^2}-\frac{\alpha}{r}$																																																									
A_4	$\mathbf{r}\cdot\mathbf{p}-i$	$\mathbf{r}\cdot\mathbf{p}-i$																																																									
Γ_4	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$																																																									
Γ_5	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$																																																									
Θ	$r(H_S-W)$	$r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$																																																									
	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$																																																									
A	$1/2m$	1																																																									
B	$-W$	m^2-E^2																																																									
C	$-\alpha$	$-2\alpha E$																																																									

Identifier	Page	Conf.	LaTeX source	Rendered	Image																								
gilmore-lie-groups_ TAB_033	290	■ 1.000	<pre>\begin{tabular}[t]{l l l l } \hline & Electric & Magnetic \\ \hline Charge density & \(\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)\) & \(\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)\) \\ \hline Current density & \(\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}\) & \(\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}\) \\ \hline Conservation law & \(\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0\) & \(\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0\) \\ \hline \end{tabular}</pre>	<table><tr><th></th><th>Electric</th><th>Magnetic</th></tr><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$</td></tr></table>		Electric	Magnetic	Charge density	$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$	<table><tr><th colspan="2">Electric</th><th>Magnetic</th></tr><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$</td></tr></table>	Electric		Magnetic	Charge density	$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$
	Electric	Magnetic																											
Charge density	$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$																											
Current density	$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$																											
Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$																											
Electric		Magnetic																											
Charge density	$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$																											
Current density	$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$																											
Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$																											

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_TAB_034	311	0.852	\begin{tabular}[t]{ l l l l l l l } \hline \multicolumn{3}{ c }{Infinitesimal generators} & \multicolumn{3}{ c }{Canonical coordinates} & Constraint \\ \hline \xi(x,y) & \eta(x,y) & \zeta(x,y,p) & R(x,y) & S(x,y) & T(x,y,p) & dS/dR \\ \hline 0 & 1 & 0 & x & y & p & T \\ 1 & 0 & 0 & y & x & p & 1/T \\ 1/a & -1/b & 0 & ax+by & bx-ay & p & (b-aT)/(a+bT) \\ x & 0 & -p & x & \ln x & xp & 1/T \\ 0 & y & p & y & \ln y & p/y & T \\ x/a & y/b & (1/b-1/a)p & y^b/x^a & b\ln y & p/x^{(a/b-1)} & (bT/R)/(bT-aR^{(1/b)}) \\ x & y & 0 & y/x & \ln y & p & (T/R)/(T-R) \\ x & -y & -2p & xy & \ln y & x^2p & (T/R)/(T+R) \\ 2x & y & -p & y^2/x & \ln y & yp & (T/R)/(2T-R) \\ x & 2y & p & y/x^2 & \ln y & p/x & (T/R)/(T-2R) \\ y & 0 & -p^2 & y & x/y & x-y/p & -T/R^2 \\ 0 & x & 1 & x & y/x & xp-y & T/R^2 \\ -y & x & 1+p^2 & \sqrt{x^2+y^2} & \tan^{-1}(y/x) & (y-xp)/(x+yp) & -T/R \\ 1 & y/x & (px-y)/x^2 & y/x & x & (xp-y)/x^2 & 1/T \\ a & x & 1 & x^2-2ay & x/a & x-ap & 1/(2aT) \\ a & y & p & x-a\ln y & x/a & p/y & (1/a)/(1-aT) \\ x & b & -p & e^y/x^b & y/b & e^y * p^b & (bR)^{-1}/[1-b(R/T)^{(1/b)}] \\ \hline y & b & -p^2 & y^2-2bx & y/b & y-b/p & 1/(2bT) \\ 0 & e^{f(x)} & f'e^f & x & y/e^f & p-yf' & T/e^{f(R)} \\ x^2 & xy & y-xp & y/x & 1/x & xp-y & 1/T \\ xy & y^2 & yp-xp^2 & y/x & 1/y & y/p-x & 1/(TR^2) \\ xy & 0 & -yp-xp^2 & y & (\ln x)/y & y/(xp)-\ln x & T/R^2 \\ 0 & xy & y+xp & x & (\ln y)/x & xp/y-\ln y & T/R^2 \\ g(y) & 0 & -g'p^2 & y & x/g & 1/p-xg'/g & T/g(R) \\ 0 & f(x) & f' & x & y/f & fp-f'y & T/f^2(R) \\ f(x) & 0 & -f'p & y & F (F'f=1) & pf & 1/T \\ 0 & g(y) & g'p & x & G (G'g=1) & p/g & T \\ x^{k+1} & kx^ky & x^k(k^2y/x-p) & y/x^k & 1/x^k & xp-ky & -k/T \\ kxy^k & y^{k+1} & y^k(p-k^2xp^2/y) & x/y^k & 1/y^k & y/p-kx & -k/T \\ \hline \end{tabular}		

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																										
gilmore-lie-groups_ TAB_035	316	■1.000	$\begin{tabular}{t}{ l l l } \hline 1830 & \text{Galois} & \text{solve algebraic equations} \\ \hline 1874 & \text{Lie} & \text{solve differential equations} \\ \hline 1969 & \text{Risch} & \text{integrate in closed form} \\ \hline ? & - & \text{solve determining equations for } \xi, \eta \\ \hline ? & - & \text{solve determining equations for } R, S, T \\ \hline ? & - & \text{solve } F(R, -, T) = 0 \text{ for } R. \\ \hline \end{tabular}$	<table><tr><td>1830</td><td>Galois</td><td>solve algebraic equations</td></tr><tr><td>1874</td><td>Lie</td><td>solve differential equations</td></tr><tr><td>1969</td><td>Risch</td><td>integrate in closed form</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for ξ, η</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for R, S, T</td></tr><tr><td>?</td><td>-</td><td>solve $F(R, -, T) = 0$ for R.</td></tr></table>	1830	Galois	solve algebraic equations	1874	Lie	solve differential equations	1969	Risch	integrate in closed form	?	-	solve determining equations for ξ, η	?	-	solve determining equations for R, S, T	?	-	solve $F(R, -, T) = 0$ for R .	<table><tr><td>1830</td><td>Galois</td><td>solve algebraic equations</td></tr><tr><td>1874</td><td>Lie</td><td>solve differential equations</td></tr><tr><td>1969</td><td>Risch</td><td>integrate in closed form</td></tr><tr><td>?</td><td>–</td><td>solve determining equations for ξ, η</td></tr><tr><td>?</td><td>–</td><td>solve determining equations for R, S, T</td></tr><tr><td>?</td><td>–</td><td>solve $F(R, -, T) = 0$ for R.</td></tr></table>	1830	Galois	solve algebraic equations	1874	Lie	solve differential equations	1969	Risch	integrate in closed form	?	–	solve determining equations for ξ, η	?	–	solve determining equations for R, S, T	?	–	solve $F(R, -, T) = 0$ for R .						
1830	Galois	solve algebraic equations																																													
1874	Lie	solve differential equations																																													
1969	Risch	integrate in closed form																																													
?	-	solve determining equations for ξ, η																																													
?	-	solve determining equations for R, S, T																																													
?	-	solve $F(R, -, T) = 0$ for R .																																													
1830	Galois	solve algebraic equations																																													
1874	Lie	solve differential equations																																													
1969	Risch	integrate in closed form																																													
?	–	solve determining equations for ξ, η																																													
?	–	solve determining equations for R, S, T																																													
?	–	solve $F(R, -, T) = 0$ for R .																																													
gilmore-lie-groups_ TAB_036	318	■1.000	$\begin{tabular}{t}{ l l l } \hline \text{\(k\)} & \text{Potential type} & \text{Transformation} \\ \hline -1 & \text{Coulomb} & \gamma^2 = \beta^3 \\ \hline 0 & \text{no force} & \gamma^2 = \beta^2 \\ \hline +1 & \text{local gravitational potential} & \gamma^2 = \beta^1 \\ \hline +2 & \text{harmonic oscillator} & \gamma^2 = \beta^0 \\ \hline \end{tabular}$	<table><tr><td>k</td><td>Potential type</td><td>Transformation</td></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$	<table><tr><td>k</td><td>Potential type</td><td>Transformation</td></tr><tr><td>−1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	−1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$												
k	Potential type	Transformation																																													
-1	Coulomb	$\gamma^2 = \beta^3$																																													
0	no force	$\gamma^2 = \beta^2$																																													
+1	local gravitational potential	$\gamma^2 = \beta^1$																																													
+2	harmonic oscillator	$\gamma^2 = \beta^0$																																													
k	Potential type	Transformation																																													
−1	Coulomb	$\gamma^2 = \beta^3$																																													
0	no force	$\gamma^2 = \beta^2$																																													
+1	local gravitational potential	$\gamma^2 = \beta^1$																																													
+2	harmonic oscillator	$\gamma^2 = \beta^0$																																													
gilmore-lie-groups_ TAB_037	319	■1.000	$\begin{tabular}{t}{ l l l } \hline v_i & \text{Infinitesimal} & e^{\epsilon v_i} f(x, t) = \\ \hline v_1 & \partial_x & f(x - \epsilon, t) \\ \hline v_2 & \partial_t & f(x, t - \epsilon) \\ \hline v_3 & u \partial_u & e^\epsilon f(x, t) \\ \hline v_4 & x \partial_x + 2t \partial_t & f(e^{-\epsilon} x, e^{-2\epsilon} t) \\ \hline v_5 & 2t \partial_x - x u \partial_u & e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t) \\ \hline v_6 & 4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u & \lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t) \text{ where } \lambda^2 = 1/(1 + 4\epsilon t) \\ \hline \end{tabular}$	<table><tr><td>v_i</td><td>Infinitesimal</td><td>$e^{\epsilon v_i} f(x, t) =$</td></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t) \text{ where } \lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t) \text{ where } \lambda^2 = 1/(1 + 4\epsilon t)$	<table><tr><td>v_i</td><td>Infinitesimal</td><td>$e^{\epsilon v_i} f(x, t) =$</td></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$																																													
v_1	∂_x	$f(x - \epsilon, t)$																																													
v_2	∂_t	$f(x, t - \epsilon)$																																													
v_3	$u \partial_u$	$e^\epsilon f(x, t)$																																													
v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$																																													
v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$																																													
v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t) \text{ where } \lambda^2 = 1/(1 + 4\epsilon t)$																																													
v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$																																													
v_1	∂_x	$f(x - \epsilon, t)$																																													
v_2	∂_t	$f(x, t - \epsilon)$																																													
v_3	$u \partial_u$	$e^\epsilon f(x, t)$																																													
v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$																																													
v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$																																													
v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$																																													
gilmore-lie-groups_ TAB_038	320	■1.000	$\begin{tabular}{t}{ l l } \hline \text{Symmetry} & \text{Conserved quantity} \\ \hline \text{Space displacements} & \text{momentum} \\ \hline \text{Time displacements} & \text{energy} \\ \hline \text{Space-time displacements} & \text{four-momentum} \\ \hline \text{Rotations} & \text{angular momentum} \\ \hline \end{tabular}$	<table><tr><td>Symmetry</td><td>Conserved quantity</td></tr><tr><td>Space displacements</td><td>momentum</td></tr><tr><td>Time displacements</td><td>energy</td></tr><tr><td>Space-time displacements</td><td>four-momentum</td></tr><tr><td>Rotations</td><td>angular momentum</td></tr></table>	Symmetry	Conserved quantity	Space displacements	momentum	Time displacements	energy	Space-time displacements	four-momentum	Rotations	angular momentum	<table><tr><td>Symmetry</td><td>Conserved quantity</td></tr><tr><td>Space displacements</td><td>momentum</td></tr><tr><td>Time displacements</td><td>energy</td></tr><tr><td>Space-time displacements</td><td>four-momentum</td></tr><tr><td>Rotations</td><td>angular momentum</td></tr></table>	Symmetry	Conserved quantity	Space displacements	momentum	Time displacements	energy	Space-time displacements	four-momentum	Rotations	angular momentum																						
Symmetry	Conserved quantity																																														
Space displacements	momentum																																														
Time displacements	energy																																														
Space-time displacements	four-momentum																																														
Rotations	angular momentum																																														
Symmetry	Conserved quantity																																														
Space displacements	momentum																																														
Time displacements	energy																																														
Space-time displacements	four-momentum																																														
Rotations	angular momentum																																														